A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

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Abstract

Minimum-mass design of steel frames is usually carried out against deterministic code checks, which leaves the reliability of the optimised frame unquantified. This paper presents an open, reproducible framework for reliability-constrained, multi-objective optimisation of planar steel moment frames built on the PyNite finite element library. Discrete W-shape sections are checked against AISC 360-16 with the Direct Analysis Method. The ultimate limit state is defined at component level: it is the first attainment of a member strength limit state in a second-order elastic analysis (B_1–B_2 amplification of superposed linear responses), not structural collapse. Fifty-year drift and deflection serviceability are treated alongside it under load, material, fabrication and imperfection uncertainty, with model-form uncertainty examined separately. A sample average approximation with common random numbers turns the reliability constraints into deterministic ones for a genetic algorithm and NSGA-II. Every optimum is re-verified with 10^6 independent samples and reported with exact binomial confidence intervals, and a code-compliant variant requires the deterministic AISC and drift checks in addition to the reliability targets. The surrogate is validated against OpenSeesPy and against full second-order re-analysis for eight designs, including targeted samples near the limit state; reliability indices differ by at most 0.07. For seven frames that vary the number of storeys, the bay width and the wind pressure, the code-compliant reliability-based optima met targets of 3.0 (ultimate) and 1.5 (serviceability) and were 1.1–10.1% heavier than the deterministic optima, mainly because the deterministic optima had low drift reliability. Where a deterministic design with a tightened drift limit (up to $h/600$) met the same targets, the reliability-based optimum was within -5.0 to $+0.6$% of the

lightest such design. Uncertainty-aware optimisation did not reduce the steel mass relative to code-based optimisation. Its benefit is a least-mass design at a stated and uniform reliability.

Keywords: Reliability-based design optimisation, Steel moment frames, Monte Carlo simulation, Direct Analysis Method, NSGA-II, Serviceability, Model uncertainty, PyNite

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1. Introduction

Minimum-weight design of steel moment frames with discrete catalogue sections is a mature topic in structural optimisation. Genetic algorithms (Camp et al., 1998; Pezeshk et al., 2000), harmony search (Degertekin, 2008) and a large family of other metaheuristics (Saka and Geem, 2013) have been used to select hot-rolled sections that satisfy code strength and serviceability checks at least weight. Almost all of this work is *deterministic*: the optimiser drives the governing load-and-resistance-factor design (LRFD) checks to unity and treats the partial factors of the code as a sufficient representation of uncertainty. The partial factors, however, were calibrated for members and typical load ratios (Ellingwood et al., 1980; Galambos et al., 1982); they do not guarantee a uniform reliability for an optimised *system*, whose members are all deliberately pushed to their limits and whose governing constraints may be strength in one member group and drift in another.

Reliability-based design optimisation (RBDO) replaces or supplements the deterministic checks by explicit constraints on failure probabilities (Frangopol, 1985; Enevoldsen and Sørensen, 1994; Valdebenito and Schuëller, 2010; Aoues and Chateauneuf, 2010). Table 1 summarises the studies closest to the present one. Reliability-based optimisation of steel frames with discrete sections and Monte Carlo simulation was pioneered by Papadrakakis, Lagaros and co-workers (Papadrakakis and Lagaros, 2002; Papadrakakis et al., 2004; Lagaros and Papadrakakis, 2007), who combined evolution strategies or NSGA-II with Monte Carlo simulation (with neural-network surrogates or Latin hypercube sampling) and found, for a seismically loaded planar frame, that the reliability-based design was heavier than the deterministic one (Papadrakakis et al., 2004). Multi-objective genetic algorithms have been used for probabilistic performance-based design of steel moment frames (Foley et al., 2007) and for weight–reliability fronts of transmission towers (Mathakari et al., 2007); nonlinear inelastic analysis has been embedded in sampling-based RBDO

of steel space frames (Truong and Kim, 2017); and the system reliability of frames designed by advanced analysis has been quantified without optimisation (Buonopane and Schafer, 2006; Zhang et al., 2016). As far as the publication records we consulted show, these studies do not (i) compare reliability-based optima with code-based optima of the same frames under a *code-calibrated* load and resistance model for wind and gravity, with and without the code checks as additional constraints, (ii) report the statistical uncertainty of the estimated failure probabilities of the final designs, (iii) validate the reliability surrogate against full second-order re-analysis in the failure tail of several designs, or (iv) release the code and raw results needed to regenerate every number.

This paper addresses the research question: *can uncertainty-aware optimisation produce lighter steel frame designs than code-based optimisation while satisfying explicit reliability and serviceability constraints?* Steel mass is the objective; it is used as a proxy for material cost, and its relation to a broader cost is examined with an illustrative cost model. The contribution is not a new optimisation algorithm, but a computational workflow, with demonstrated accuracy, that makes probabilistic constraints practical in discrete steel-section optimisation and allows an honest comparison with code-based design. Specifically:

1. an exact superposition surrogate that turns five linear analyses per design (PyNite (Brinck, 2026)) and the AISC 360-16 B_1–B_2 amplification into a vectorised Monte Carlo evaluation of a series-system ultimate limit state (ULS) and two serviceability limit states (SLS) for 5×10^4–10^6 samples per design, combined with a sample average approximation (SAA) (Kleywegt et al., 2002) that makes the reliability constraints deterministic functions of the design;
2. four formulations solved with the same optimiser on the same frames: deterministic design optimisation (DDO) with several drift limits, RBDO with reliability constraints only, a *code-compliant* RBDO (RBDO-C) that also enforces every LRFD and serviceability check, and a bi-objective mass–reliability formulation;
3. a statistical treatment of the reliability estimates (exact binomial confidence intervals, failure counts, an explicit acceptance rule, and a confidence-bounded variant of the reliability constraint), and an assessment of optimiser robustness by repeated runs, SAA variants and exhaustive enumeration of a reduced design space;

4. a verification chain from closed-form solutions and an independent solver (OpenSeesPy (McKenna et al., 2010; Zhu et al., 2018)) to full P-Δ reanalysis Monte Carlo of eight designs from six frame configurations, including the failure tail and the failure classification; and
5. an open release of the code, input data, seeds and raw results with a one-command reproduction.

What is demonstrated is restricted to regular planar steel moment frames of three to nine storeys under gravity and wind, with a second-order elastic, component-based ULS. Generality to three-dimensional frames, seismic actions, inelastic collapse reliability and full cost models is *not* demonstrated here.

Section 2 presents the frame family, the design checks, the probabilistic model, the reliability estimation and the optimisation formulations. Section 3 verifies the structural and reliability models. Section 4 reports the optimiser robustness, the comparison of deterministic and reliability-based designs, the cost–reliability trade-off and the parametric, economic and sensitivity studies. Section 5 discusses the interpretation and limitations and Section 6 concludes.

2. Framework

2.1. Parameterised frame family

The framework generates regular planar moment frames with n_s storeys and n_b bays of width b (Fig. 1). All beam–column connections are rigid and column bases are fixed. The ground storey is 4.5 m high and the upper storeys 3.8 m. Frames are spaced at 7.5 m, which defines the tributary width of floor loads and wind pressure. Nominal floor loads are $q_D = 4.0$ kPa (slab, finishes and services) and $q_L = 2.4$ kPa (office occupancy); the roof carries 3.0 kPa dead and 1.0 kPa live load. Member self-weight is added to the dead load. The nominal 50-year wind is represented by a net (windward plus leeward) pressure p_w, converted to storey forces from the tributary storey heights and distributed equally to the floor nodes, which mimics a rigid floor diaphragm.

Members are grouped by storey groups of two storeys (three for the nine-storey frame); every group has three discrete design variables: the exterior column, the interior column and the beam section. Columns are selected from 20 W14 shapes (W14×48 to W14×311) and beams from 28 W16–W30 shapes. All sections are ASTM A992 ($F_y = 345$ MPa, $E = 200$ GPa). Section properties were taken from the AISC Shapes Database (American Institute

Table 1: Closely related studies. Entries are taken from the publications' abstracts and bibliographic records consulted by the authors; n.r. = not reported there; – = not applicable. MRF: moment-resisting frame; MCS: Monte Carlo simulation; LHS: Latin hypercube sampling; ES: evolution strategies; FR/PR: fully/partially restrained connections.

Study	Structure and analysis	Design variables	Reliability formulation	Optimiser	Independent verification	Public code/data
Papadrakakis and Lagaros (2002)	Large-scale structures; elastoplastic analysis to collapse	n.r.	MCS with importance sampling; plastic collapse	ES; neural-network surrogate of constraints or analysis	n.r.	n.r.
Papadrakakis et al. (2004)	3-storey, 4-bay planar steel MRF; response-spectrum analysis	Discrete W-shapes, 5 groups	MCS (LHS); system, drift $>$ 4% of storey height; $P_f \leq 10^{-3}$	$(\mu + \lambda)$-ES	n.r.	n.r.
Lagaros and Papadrakakis (2007)	Steel buildings, seismic	n.r.	MCS; robust (two-objective) and reliability-based formulations	Modified NSGA-II	n.r.	n.r.
Foley et al. (2007)	Single- and multi-storey planar steel MRFs with FR/PR connections, seismic	n.r.	Probabilistic performance-based (life safety, damage)	Multi-objective GA	n.r.	n.r.
Mathakari et al. (2007)	3D trusses (transmission towers); FE	Topology, geometry, member sizes	FE reliability, importance sampling; deformation and stress thresholds under wind	MOGA (weight vs. β)	n.r.	n.r.
Buonopane and Schafer (2006)	Planar steel frames; advanced (inelastic) analysis	– (assessment)	System reliability of frames designed by advanced analysis	–	–	n.r.
Zhang et al. (2016)	Planar steel frames; advanced analysis	– (assessment)	System-reliability framework for system-based design	–	–	n.r.
Truong and Kim (2017)	Steel space frames; practical advanced (nonlinear inelastic) analysis	Discrete	Improved LHS and importance sampling	Harmony search	n.r.	n.r.
This study	Planar steel MRFs; second-order elastic (DAM, B_1–B_2 surrogate of five linear analyses)	Discrete AISC W-shapes, 9 variables	Crude MCS with SAA; series-system first-member ULS (two Turkstra scenarios), drift and deflection SLS; with and without the LRFD checks; exact binomial CIs	GA + local search; NSGA-II; exhaustive enumeration of a reduced space	Closed form; OpenSeesPy; full P-Δ MC of 8 designs incl. tail strata	Yes: code, inputs, raw results, seeds

of Steel Construction, 2017); as a guard against transcription errors, each entry was checked against the fillet-free three-plate idealisation of the section, giving ratios of tabulated to plate values of 0.99–1.03 for A, I_x, Z_x and I_y in all cases. The catalogues are sorted by area so that an index step is a monotonic change in mass. A constructability constraint requires column areas not to increase with height.

The base frame has six storeys, three 8-m bays and $p_w = 1.2$ kPa. The parametric family varies one parameter at a time (Table 2): $n_s \in \{3, 6, 9\}$, $b \in \{6, 8, 10\}$ m and $p_w \in \{0.8, 1.2, 1.6\}$ kPa. A reduced problem, the base frame with one section group over all six storeys (three variables, $20 \times 20 \times 28 = 11\,200$ designs), is used to check the optimiser against exhaustive enumeration.

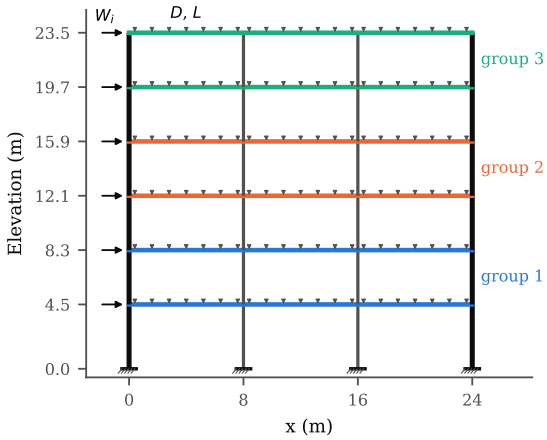


Figure 1: Base frame (six storeys, three 8-m bays). Thick lines: exterior columns; colours indicate the three storey groups, each with an exterior-column, interior-column and beam design variable. Gravity loads act on the beams; wind storey forces W_i are applied at the floor levels.

2.2. Structural analysis by superposition of PyNite unit cases

For each candidate design $\mathbf{x}$, PyNite (Brinck, 2026) solves five first-order linear elastic load cases with nominal stiffness: dead load D (including self-weight), live load L, wind W, and two lateral cases N_D and N_L whose floor forces equal the dead and live gravity load of each floor. Any realisation of the loading, with intensities (d, ℓ, w) (multiples of the nominal D, L, W) and

Table 2: Frames of the parametric study (one parameter varied at a time from the base frame).

Frame	Storeys	Bays	Bay (m)	p_w (kPa)	Groups	Variables
3 storeys	3	3	8	1.2	2	6
Base	6	3	8	1.2	3	9
9 storeys	9	3	8	1.2	3	9
Bay 6 m	6	3	6	1.2	3	9
Bay 10 m	6	3	10	1.2	3	9
$p_w = 0.8$ kPa	6	3	8	0.8	3	9
$p_w = 1.6$ kPa	6	3	8	1.6	3	9

a global out-of-plumbness angle θ, then produces the first-order response

$$r = \underbrace{d\, r_D + \ell\, r_L}_{r_{nt}} + \underbrace{w\, r_W + \theta\,(d\, r_{N_D} + \ell\, r_{N_L})}_{r_{lt}}, \tag{1}$$

split into the no-translation part r_{nt} and the lateral-translation part r_{lt} required by the approximate second-order analysis of AISC 360 Appendix 8. Because member forces of a frame with uniformly scaled modulus eE are independent of e and displacements scale with $1/e$, a global random modulus factor is also represented exactly. Responses extracted per unit case are column end moments, axial and shear forces; beam moments at 11 stations along the span, axial and end shear forces; mid-span beam deflections relative to the chord; and floor-average inter-storey drifts.

Second-order effects are introduced with the B_1–B_2 method. For storey i,

$$B_{2,i} = \left(1 - \frac{P_{story,i}}{P_{e,story,i}}\right)^{-1}, \qquad P_{e,story,i} = R_M\, \tau\, e\, \frac{H_i h_i}{\Delta_{H,i}}, \tag{2}$$

where $P_{story,i}$ is the total gravity load carried by the storey, $H_i/\Delta_{H,i}$ the first-order lateral stiffness obtained from the W case, h_i the storey height and $R_M = 0.85$. For columns, $B_1 = C_m/(1 - P_r/P_{e1}) \geq 1$ with $C_m = 0.6 - 0.4 M_1/M_2$ computed from the no-translation end moments and $P_{e1} = \pi^2 \tau e EI/L^2$. Required strengths are $M_r = B_1 M_{nt} + B_2 M_{lt}$ and $P_r = P_{nt} + B_2 P_{lt}$; beams use the larger B_2 of the two storeys they connect. A storey with $P_{story} \geq P_{e,story}$ is unstable and is treated as failure.

The amplifiers are *recomputed for every sample*: P_{story}, P_r and C_m follow the sampled gravity loads, and $P_{e,story}$ and P_{e1} follow the sampled modulus

factor e. The assumptions behind this use of the method are that (a) the lateral stiffness of a storey scales with the frame-wide modulus factor and is otherwise that of the nominal sections, (b) the gravity load is distributed over the columns as in the first-order analysis, (c) member stiffness is not reduced by yielding before the first member limit state is reached, and (d) the storey amplification is the same for all columns of a storey (rigid floor). Assumptions (a) and (b) are removed in the full re-analysis of Section 3.3, where every member has its own modulus and property factor and the P-Δ (and, through the element geometric stiffness, P-δ) effects are computed directly; assumption (c) is inherent in the elastic ULS definition below. In deterministic design, the Direct Analysis Method (DAM) (American Institute of Steel Construction, 2016; Surovek-Maleck and White, 2004) is followed: stiffness is reduced by $\tau = 0.8$, notional loads of $0.002Y_i$ are applied in gravity-only combinations and an additional $0.001Y_i$ in all combinations so that $\tau_b = 1$ may be used; effective length factors are then $K = 1$. In the reliability analysis the best-estimate stiffness ($\tau = 1$) is used and the notional loads are replaced by the random out-of-plumbness θ.

2.3. Member resistances and limit states

Nominal resistances follow AISC 360-16: compression from Chapter E3 with $K = 1$, strong-axis buckling over the storey height and weak-axis buckling between floors (columns) or between secondary beams at third points (beams); flexure from F2 (lateral–torsional buckling, $C_b = 1$) and F3 (flange local buckling); shear from G2.1; and combined forces from the H1-1 interaction equations. Beam axial forces are small (largest beam P_r/P_c of the optimised designs below 0.07), so the slender-web reduction of beam compression capacity is not modelled. For the deterministic checks the LRFD resistance factors $\phi_c = \phi_b = \phi_t = 0.9$ and $\phi_v = 1.0$ or 0.9 (G2.1) are applied to the ASCE 7-05 (American Society of Civil Engineers, 2006) strength combinations $1.4D$, $1.2D + 1.6L$, $1.2D + 0.5L \pm 1.6W$ and $0.9D \pm 1.6W$ (roof live load is treated as part of L). Serviceability requires inter-storey drift $\leq h/400$ under $D + 0.5L + W$ and beam live-load deflection $\leq$ span/360.

For each realisation the member utilisation U_k is the larger of the H1-1 interaction ratio and the shear ratio, both evaluated with nominal (unfactored) resistances that depend on the random yield stress, modulus and fabrication

factor of the member. Three limit-state functions are defined:

$$g_{ULS} = 1 - \max_{s\in\{S_1,S_2\}} \max_k U_k^{(s)}, \tag{3}$$

$$g_{drift} = 1 - \max_i \frac{\delta_i^{(S_2)}/h_i}{1/400}, \tag{4}$$

$$g_{defl} = 1 - \max_j \frac{\delta_{L,j}}{\ell_j/360}. \tag{5}$$

The load scenarios are constructed with Turkstra's rule (Turkstra and Madsen, 1980): $S_1 = D + L_{50}$ (50-year maximum live load) and $S_2 = D + L_{apt} + W_{50}$ (arbitrary-point-in-time live load with the 50-year maximum wind); the out-of-plumbness acts in both. The ULS event is the union of the member events over all members and both scenarios. It is constructed sample by sample, so the dependence between members (common loads, common modulus, partly correlated yield stress) and between the scenarios (common D, E, F_y, fabrication factor and θ) is represented exactly by the joint sampling; no bounds or independence assumptions of series-system theory are needed. Turkstra's rule itself neglects the probability that the 50-year live load and the 50-year wind coincide, which is small for the statistics used. The drift SLS is evaluated in S_2 and the deflection SLS under L_{50}.

g_{ULS} is a *first-member-failure, second-order elastic, component-based* limit state: the frame is deemed to fail when the first member reaches a code-format resistance, evaluated with the AISC interaction equations and nominal (unfactored) resistances. It is not a collapse analysis. It does not credit inelastic redistribution, and residual stresses and member out-of-straightness enter only through the column-curve and LTB equations of AISC 360, not through an explicit imperfect model. Connections are rigid and do not fail. Throughout this paper β_{ULS} denotes the reliability index of *this* event. It is consistent with the member-based calibration of current codes (Ellingwood et al., 1980), but it is a lower bound-type measure of the collapse reliability, which inelastic system analyses report to be higher (Buonopane and Schafer, 2006; Zhang et al., 2016).

2.4. Probabilistic model

Table 3 lists the basic random variables, their sources, and which parameters are assumptions of this study. The load statistics are those of the probability-based load criterion of Ellingwood et al. (1980) for dead load, 50-year live load and 50-year wind, together with the arbitrary-point-in-time live

load (gamma, mean $0.24L_n$, c.o.v. 0.60) and the fabrication (geometric) factor (c.o.v. 0.05). These statistics were used to calibrate the LRFD load factors, so using them makes the comparison between code-based and reliability-based designs internally consistent. The c.o.v. of yield stress (0.07) and modulus (0.03) and the lognormal model are those of the JCSS Probabilistic Model Code (Joint Committee on Structural Safety, 2001). The mean-to-nominal ratio of F_y (1.10) and the inter-member equicorrelation of $\ln F_y$ ($\rho = 0.5$) are assumptions; both are varied in the sensitivity study. The initial out-of-plumbness is normal with zero mean and standard deviation 1/1000, so that the erection tolerance of 1/500 corresponds to two standard deviations.

Dependence. The load intensities are fully correlated over the frame, i.e. one realisation of D, L and W scales the nominal load pattern. For dead load and wind this is the usual assumption; for live load it neglects the spatial variability between floors and bays and is conservative for the frame-level effects that govern here. The modulus factor is frame-wide in the surrogate (members from one mill order) and member-specific in the full re-analysis; the yield stress and fabrication factor are member-specific, the former with partial correlation to represent a common heat or producer. θ is a single frame-wide sway imperfection.

Aleatory versus epistemic uncertainty. The variables of Table 3 describe physical (aleatory) variability, with their parameters fixed at the cited values. Model-form (epistemic) uncertainty of the structural and resistance models, i.e. the error of the H1-1 interaction equations and of the second-order elastic analysis, is *not* part of the baseline model, so that the baseline indices are conditional on the structural model, as in the code calibration that is the reference of the comparison. Its effect is quantified separately by frame-wide lognormal model factors with unit mean on resistance (θ_R), load effect (θ_E) and drift (θ_δ), with c.o.v. of 0.05 and 0.10 chosen to bracket plausible values (Section 4.7). Because H1-1 and the shear check are homogeneous of degree one in the demands and of degree minus one in the resistances, the factors act on the utilisation exactly as $U\theta_E/\theta_R$. Parameter uncertainty of the load statistics is addressed by the sensitivity study.

2.5. Reliability estimation, confidence intervals and acceptance

Failure probabilities are estimated by crude Monte Carlo simulation. With k failures in N independent samples, the point estimate is $\hat{P}_f = k/N$, its c.o.v. is $\sqrt{(1-P_f)/(NP_f)}$, and the reported index is $\hat{\beta} = -\Phi^{-1}(\hat{P}_f)$. Because k

Table 3: Basic random variables (50-year reference period), source of each model, and which parameters are assumptions of this study.

Variable	Distribution	Mean/nominal	c.o.v.	Basis	Status
Dead load D	Normal	1.05	0.10	Ellingwood et al. (1980)	sourced
50-yr max. live load L_{50}	Gumbel	1.00	0.25	Ellingwood et al. (1980)	sourced
Arbitrary-point-in-time live load L_{apt}	Gamma	0.24	0.60	Ellingwood et al. (1980)	sourced
50-yr max. wind load W_{50}	Gumbel	0.78	0.37	Ellingwood et al. (1980)	sourced
Elastic modulus E	Lognormal	1.00	0.03	JCSS (2001): c.o.v., lognormal	sourced
Yield stress F_y (per member)	Lognormal	1.10	0.07	JCSS (2001): c.o.v., lognormal; mean ratio and ρ	assumed (mean, ρ)
Fabrication factor F (per member)	Normal	1.00	0.05	Ellingwood et al. (1980)	sourced
Out-of-plumbness θ (rad)	Normal	0	s.d. 1/1000	1/500 tolerance taken as 2 s.d.	assumed
Model factors $\theta_R, \theta_E, \theta_\delta$ (frame-wide)	Lognormal	1.00	0 (0.05, 0.10)	sensitivity only	assumed

is binomial, an exact (Clopper–Pearson) two-sided 95% confidence interval $[P_f^{lo}, P_f^{hi}]$ is computed from the beta distribution,

$$P_f^{lo} = B^{-1}(0.025;\, k,\, N-k+1)\,, \qquad P_f^{hi} = B^{-1}(0.975;\, k+1,\, N-k)\,, \quad (6)$$

and mapped to the interval $[-\Phi^{-1}(P_f^{hi}),\, -\Phi^{-1}(P_f^{lo})]$ of β. The one-sided 95% lower confidence bound of β follows from the one-sided upper bound $B^{-1}(0.95;\, k+1,\, N-k)$ of P_f. When no failure is observed, the point estimate is undefined and only the lower confidence bound is reported (for $N = 10^6$, $P_f \le 3.0 \times 10^{-6}$ and $\beta \ge 4.53$ with 95% confidence). Every final design is reported with N, k, the point estimate and the interval.

The limit states are evaluated by the superposition (1) and closed-form resistances, so a sample costs a few vectorised arithmetic operations per member; a design is evaluated with 5×10^4 samples in about one second on one CPU core, about four times the cost of the five PyNite analyses.

Sample average approximation. Inside the optimiser the samples are drawn once and reused for every design (common random numbers). This SAA (Kleywegt et al., 2002) makes $\hat{\beta}(\mathbf{x})$ a deterministic function of the design, so the comparison of two designs is not corrupted by independent sampling noise (their estimates are positively correlated) and the discrete search is well posed. The price is a *finite-sample misclassification* risk: a design whose true index is just below the target may be accepted because the particular sample is favourable, and the optimiser, which seeks the lightest accepted design, preferentially finds such designs (the optimisation bias of SAA). We control this risk in three ways: (i) every final design is re-verified with $N = 10^6$ *independent* samples; (ii) the SAA problem is re-solved with other sample streams and sizes ($N = 2 \times 10^4$, 5×10^4, 10^5) to measure the spread of the optima; and (iii) a confidence-bounded variant imposes the targets on the one-sided 95% lower confidence bound of β in the SAA sample instead of

the point estimate. To save time, a design is first evaluated on the first 10^4 samples and the remainder is computed only if the screening estimate is not clearly infeasible (more than 0.4 below a target index).

Acceptance rule. A design is *accepted* by the optimiser when its SAA estimates meet the targets (point estimates, or lower confidence bounds in the confidence-bounded variant). In the independent verification, a design is classified as meeting a target with 95% confidence ("above") when the lower end of the interval exceeds the target, as statistically indistinguishable from it ("within") when the interval contains the target, and as falling short ("below") when the upper end is smaller. A "within" result is accepted as meeting the target to the stated sampling resolution; a "below" result would require re-optimisation with a margin.

2.6. Optimisation formulations

The objective is the steel mass $m(\mathbf{x}) = \rho_s \sum_k A_k(\mathbf{x})\ell_k$, used as a proxy for material cost (connection, fabrication and erection costs are considered only in the illustrative cost model of Section 2.7). The formulations are

$$\text{DDO-}h/n: \quad \min_{\mathbf{x}} \; m \quad \text{s.t.} \quad \max_{c,k} U_{c,k}^{LRFD} \le 1, \quad \frac{\delta}{h} \le \frac{1}{n}, \quad \delta_L \le \frac{\ell}{360}, \tag{7}$$

$$\text{RBDO:} \quad \min_{\mathbf{x}} \; m \quad \text{s.t.} \quad \hat{\beta}_{ULS} \ge \beta_T, \quad \hat{\beta}_{drift} \ge \beta_{SLS}, \quad \hat{\beta}_{defl} \ge \beta_{SLS}, \tag{8}$$

$$\text{RBDO-C:} \quad \min_{\mathbf{x}} \; m \quad \text{s.t.} \quad \text{the constraints of DDO-}h/400 \text{ and of RBDO}, \tag{9}$$

$$\text{MO:} \quad \min_{\mathbf{x}} \; \left(m, \, -\hat{\beta}_{ULS}\right) \quad \text{s.t.} \quad \hat{\beta}_{drift}, \hat{\beta}_{defl} \ge \beta_{SLS}, \quad \hat{\beta}_{ULS} \ge 1, \tag{10}$$

all subject to column areas that do not increase with height. DDO with $n = 400$ ("DDO") is the code-based design; $n = 450$, 500 and 600 represent the range of tighter drift limits used in practice. RBDO-C is the formulation that a designer bound by the code could adopt: the reliability constraints are added to, not substituted for, the mandatory checks, so that every RBDO-C design is code-compliant and its mass is directly comparable with DDO. RBDO without the code checks is retained to show what the reliability constraints alone would select; a design accepted by RBDO but failing an LRFD check would have no regulatory standing under a prescriptive code, although it could be admissible under a reliability-based (performance-based) route of the kind allowed by EN 1990 (CEN, 2002).

The baseline targets are $\beta_T = 3.0$ for the ULS, the value adopted by Ellingwood et al. (1980) for gravity load combinations in the calibration of the LRFD load factors (a lower value was accepted for combinations with wind), and $\beta_{SLS} = 1.5$ (50 years), the value given by EN 1990 (CEN, 2002) for irreversible serviceability states. Because both targets are choices of the owner or code writer, results are also given for other targets.

DDO and RBDO(-C) are solved with the same elitist genetic algorithm (population 40, binary tournament with Deb's feasibility rules (Deb, 2000), uniform crossover with probability 0.9, per-gene mutation probability $1.5/n_v$ of which 70% are steps of ± 1–3 catalogue positions and 30% random resets, and $(\mu + \lambda)$ replacement). A run stops after 15 generations without improvement of the best design or after 60 generations. The best GA design is then polished by a discrete local search that accepts the first feasible lighter neighbour among single down-sizing moves and down/up-sizing swaps of two variables, repeated until no improvement is found, so every reported optimum is locally optimal with respect to these moves. The GA is a heuristic: convergence of repeated runs indicates robust convergence within the search budget, not a guarantee of global optimality, which is checked only for the reduced problem by enumeration. The bi-objective problem is solved with NSGA-II (Deb et al., 2002) with constrained domination (population 48, 60 generations); all feasible designs visited are archived and the non-dominated archive is re-verified. The SAA samples, the GA operators and the verification samples are all seeded, so every number in this paper is reproducible from the released code.

2.7. Illustrative cost model

To test whether conclusions drawn from steel mass carry over to cost, a simple cost index in tonnes of steel equivalent is used,

$$C = m + \tau\, n_{sec} + \kappa\, m_{beam}, \tag{11}$$

where n_{sec} is the number of distinct section designations (procurement, detailing and handling), m_{beam} the beam mass (a proxy for the size of moment connections and their welding, which scale with the beam flange force), and τ and κ are cost ratios. The number of members, connections and pieces is identical for all designs of a frame and therefore does not change the comparison. Equation (11) is not a calibrated cost model; it is used only to show how large the non-mass terms would have to be to change the ranking of the designs.

3. Verification of the structural and reliability models

The conclusions of an optimisation study are only as credible as the models it optimises. Four verification levels were therefore applied before any design comparison was made.

3.1. Closed-form benchmarks

Table 4 compares PyNite with closed-form solutions for a cantilever, a fixed–fixed beam under uniform load, and a fixed-base portal frame under lateral load. The portal solution follows from the slope-deflection method for inextensible members, with $k = (I_b/L)/(I_c/h)$,

$$\Delta = \frac{Hh^3(2+3k)}{12EI_c(1+6k)}, \qquad M_{base} = \frac{Hh(1+3k)}{2(1+6k)}, \qquad M_{top} = \frac{3kHh}{2(1+6k)}. \quad (12)$$

Agreement is at round-off level for the statically simple cases and within 10^{-5} for the portal, the residual being the axial flexibility that PyNite includes. The second-order response of a cantilever beam-column with axial load P and tip load H has the exact tip sway $\Delta = \frac{H}{Pk}(\tan kL - kL)$, $k = \sqrt{P/EI}$. PyNite's P-Δ solver with a single element is within 0.6% of the exact value even at $P = 0.6P_{cr}$, and converges to $< 2\times10^{-6}$ with eight elements (Table 4), showing that its geometric stiffness also captures the member (P-δ) effect.

Table 4: PyNite versus closed-form solutions ($E = 200$ GPa).

Quantity	Closed form	PyNite	Rel. error
Cantilever tip deflection (m)	0.053333	0.053333	0.0e+00
Cantilever fixed-end moment (kN m)	200	200	0.0e+00
Fixed beam end moment (kN m)	213.33	213.33	4.0e-16
Fixed beam mid-span moment (kN m)	106.67	106.67	1.1e-15
Fixed beam mid-span deflection (m)	0.010667	0.010667	3.3e-16
Portal sway (m)	0.012698	0.012699	9.3e-06
Portal column base moment (kN m)	114.29	114.29	7.2e-06
Portal column top moment (kN m)	85.714	85.715	3.6e-06
Second-order cantilever, tip sway (mm): exact vs PyNite P-Δ with n elements			
P/P_{cr}	exact	$n=1$ / $n=2$ / $n=8$	err. ($n=1$, $n=8$)
0.2	25.972	25.964 / 25.972 / 25.972	3.3e-04, 1.0e-07
0.4	34.534	34.471 / 34.529 / 34.534	1.8e-03, 5.5e-07
0.6	51.650	51.326 / 51.627 / 51.650	6.3e-03, 1.8e-06

3.2. Independent solver

The optimised designs were rebuilt in OpenSeesPy (McKenna et al., 2010; Zhu et al., 2018) with elastic beam-column elements, element loads for the distributed gravity and self-weight and nodal storey forces, independently of the PyNite model. Table 5 lists the largest relative difference of each response quantity, normalised by the largest absolute value of that quantity in the frame. For all five unit cases the two solvers agree to 10^{-13}–10^{-16}, i.e. to machine precision. (Gravity-load drifts of these symmetric frames are themselves of the order of round-off and are not compared.) For the P-Δ analysis of the combination $1.2D + 0.5L + 1.6W$, OpenSees used the `PDelta` coordinate transformation and a load-controlled Newton solution. PyNite's second-order drifts differ from OpenSees by at most 0.8%, and the B_1–B_2 amplified first-order drifts used by the surrogate by at most 1.8% (Fig. 2). The largest differences occur in the upper storeys, where the drifts and hence the governing drift checks are small; in the lower storeys, which govern the drift limit state, the surrogate exceeds the P-Δ solution by up to 1.8%, and it underestimates it by at most 1.0% in any storey. Column base moments behave similarly (at most 1.6%). These differences are an order of magnitude smaller than the sampling resolution of the reliability constraints at the SAA sample size (Section 3.4).

3.3. Reliability surrogate versus full second-order re-analysis

The surrogate rests on three simplifications: second-order effects through B_1–B_2, a modulus that is uniform over the frame, and nominal section stiffness (the fabrication factor enters the resistance only). In the full model, every sample is analysed with a PyNite P-Δ analysis in which every member receives its own modulus factor (c.o.v. 0.03) and its fabrication factor is applied to A and I; the surrogate is evaluated on the same samples with the mean of the member modulus factors. Because agreement of mean responses does not guarantee agreement of small failure probabilities, the comparison was designed around the failure classification and the tail, for eight final designs from six frame configurations (the base DDO design and the RBDO-C designs of all seven frames, including nine storeys, 10-m bays and high wind):

- a *random set* of 600 samples of scenario S_2, which compares utilisation and drift and, because $\beta_{drift} \approx 1.5$ for these designs, the drift classification at the actual SLS constraint level;

Table 5: Maximum relative differences between PyNite and OpenSeesPy for the optimised base designs and two other frames: linear unit cases and P-Δ analysis of $1.2D+0.5L+1.6W$.

Design	Case	Drift	Column M	Column N	Beam end M
Base DDO	D	–	8.3e-16	2.0e-15	8.0e-16
	L	–	4.7e-16	1.6e-15	9.6e-16
	W	1.8e-14	1.1e-14	9.8e-15	1.4e-14
	ND	1.8e-14	1.3e-14	8.2e-15	1.4e-14
	NL	1.5e-14	1.0e-14	8.7e-15	1.3e-14
	P-Δ: PyNite	7.3e-03	7.4e-03		
	P-Δ: B_1–B_2	1.8e-02	1.6e-02		
Base RBDO-C	D	–	6.7e-16	1.4e-15	4.7e-16
	L	–	6.7e-16	1.3e-15	6.7e-16
	W	4.2e-14	4.3e-14	3.0e-14	4.2e-14
	ND	4.2e-14	4.2e-14	2.9e-14	4.1e-14
	NL	4.2e-14	4.3e-14	3.0e-14	4.2e-14
	P-Δ: PyNite	7.1e-03	6.2e-03		
	P-Δ: B_1–B_2	1.5e-02	1.5e-02		
3-storey DDO	D	–	3.1e-16	1.1e-15	4.9e-16
	L	–	7.7e-16	8.8e-16	5.5e-16
	W	6.0e-15	6.9e-15	2.8e-15	4.0e-15
	ND	6.0e-15	6.8e-15	2.9e-15	4.4e-15
	NL	6.8e-15	7.4e-15	3.5e-15	4.4e-15
	P-Δ: PyNite	5.7e-03	6.2e-03		
	P-Δ: B_1–B_2	9.4e-03	1.5e-02		
9-storey DDO	D	–	9.5e-16	2.0e-15	1.1e-15
	L	–	1.4e-15	2.8e-15	2.0e-15
	W	1.7e-13	1.1e-13	1.3e-13	1.4e-13
	ND	1.7e-13	1.2e-13	1.3e-13	1.4e-13
	NL	1.7e-13	1.2e-13	1.3e-13	1.4e-13
	P-Δ: PyNite	8.1e-03	7.5e-03		
	P-Δ: B_1–B_2	1.8e-02	1.5e-02		

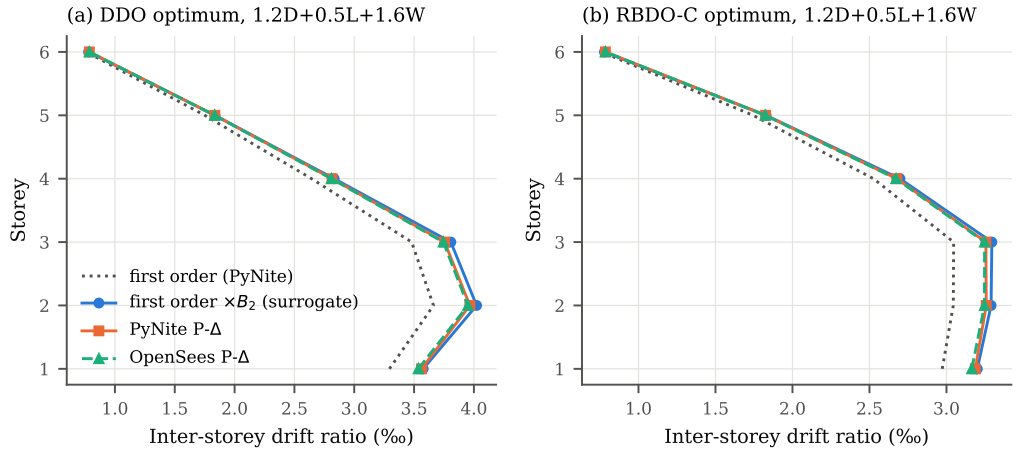


Figure 2: Inter-storey drift profiles of the base designs under $1.2D+0.5L+1.6W$: first-order PyNite, the B_2-amplified surrogate, PyNite P-Δ and OpenSees P-Δ. PyNite and OpenSees P-Δ differ by at most 0.8%. The surrogate differs from them by at most 1.8%, with the largest relative differences in the upper storeys; it underestimates the drift by at most 1.0%.

- a *tail-stratified set* drawn from a pool of 2×10^5 samples: stratum A contains all samples the surrogate classifies as ULS failures ($U_s \geq 1$; up to 250 analysed), and stratum B the near misses $0.85 \leq U_s < 1$ (250 analysed at random). Both scenarios are re-analysed when their surrogate utilisation is in the tail. With the stratum probabilities known from the pool, the full-model failure probability is estimated as $\hat{P}_f^{full} = P(A)\hat{f}_A + P(B)\hat{f}_B$, where $\hat{f}$ is the fraction of full-model failures in a stratum. Samples with $U_s < 0.85$ cannot fail in the full model, since the largest observed ratio U_{full}/U_s over all 8 000 analysed samples is 1.049. A 95% interval of $\hat{P}_f^{full}$, of the ratio $\hat{P}_f^{full}/\hat{P}_f^{sur}$ and of the difference in β is obtained by propagating Jeffreys posteriors of the stratum fractions and of the pool counts by simulation.

The base RBDO-C design was in addition analysed under doubled wind (1000 samples) as a stress test that produces failure fractions large enough to be compared directly.

Table 6 and Fig. 3 summarise the results. Over all designs the surrogate overestimates the system utilisation by 0.4–1.1% on average (standard deviation of the difference 1.0–1.6%) and is unbiased for drift (−1.0 to +1.4%, standard deviation 1.9–3.1%). In the random sets, 0 to 7 of 600 drift classi-

fications differ (at most 1.2%), and the numbers of drift exceedances agree within their sampling scatter. In the tail, 79–91% of the surrogate failures are also failures of the full model and at most 4 of 250 near misses fail in the full model. The resulting full-model failure probabilities differ from the surrogate ones by $\Delta\beta_{ULS}$ between -0.07 and $+0.05$; for seven of the eight designs the 95% interval of $\Delta\beta$ contains zero. The exception is the three-storey frame, for which the full model gives $\hat{P}_f^{full} = 1.64 \times 10^{-3}$ against 1.31×10^{-3} ($\Delta\beta = -0.07$, 95% interval $[-0.17, -0.01]$): in this strength-governed, low-rise frame, four near misses fail in the full model, so the surrogate is slightly unconservative. Under doubled wind (base design), the failure fractions are 2.1% (surrogate) and 1.7% (full analysis), and the drift exceedance fractions 68.1% and 67.8%.

In summary, the surrogate reproduces the full second-order reliability within about ± 0.07 in β_{ULS}, which is of the order of the SAA resolution, and the drift reliability without detectable bias. Where it errs, it errs on the conservative side in six of eight designs. The three-storey result shows that the agreement is not guaranteed: a design accepted at the target by the surrogate may have a full-model index up to about 0.1 lower, and a validation of this kind should accompany any application of the surrogate to frame types outside the tested range.

Table 6: Surrogate versus full P-Δ re-analysis with member-by-member random stiffness and section properties. Random set: 600 samples of S_2 (bias = mean ± standard deviation of surrogate/full -1; k_{drift} = drift exceedances; discordant = samples classified differently). Tail strata (pool 2×10^5): full-model failures / samples analysed. $\hat{P}_f$ from the pool (surrogate) and the stratified estimator (full model); intervals from Jeffreys posteriors.

	Random set, scenario S_2				Tail strata: full failures				
Design	U bias (%)	drift bias (%)	k_{drift} full/sur.	discordant	A: $U_s \geq 1$	B: $0.85 \leq U_s < 1$	$\hat{P}_f^{sur}$	$\hat{P}_f^{full}$ [95% CI]	$\Delta\beta_{ULS}$ [95% CI]
Base DDO	+1.1 ± 1.2	+1.0 ± 2.2	96/103	7	203/233	1/250	1.17e-03	1.11e-03 [9.46e-04, 1.47e-03]	+0.02 [-0.07, +0.04]
Base RBDO-C	+0.8 ± 1.1	-0.6 ± 2.2	43/37	6	196/220	1/250	1.10e-03	1.07e-03 [9.12e-04, 1.42e-03]	+0.01 [-0.07, +0.04]
3 storeys RBDO-C	+0.4 ± 1.6	-0.1 ± 3.1	2/2	0	227/250	4/250	1.31e-03	1.64e-03 [1.31e-03, 2.27e-03]	-0.07 [-0.17, -0.01]
9 storeys RBDO-C	+0.8 ± 1.1	-1.0 ± 1.9	38/38	4	54/68	1/250	3.40e-04	3.07e-04 [2.31e-04, 4.58e-04]	+0.03 [-0.07, +0.08]
Bay 6 m RBDO-C	+1.0 ± 1.3	-0.5 ± 2.2	36/35	1	17/21	0/250	1.05e-04	8.50e-05 [5.30e-05, 1.38e-04]	+0.05 [-0.02, +0.11]
Bay 10 m RBDO-C	+1.1 ± 1.0	-0.4 ± 2.5	27/28	1	142/171	1/250	8.55e-04	7.87e-04 [6.53e-04, 1.09e-03]	+0.02 [-0.07, +0.06]
$p_w = 0.8$ kPa RBDO-C	+0.8 ± 1.1	+1.4 ± 2.5	29/31	4	197/221	2/250	1.11e-03	1.20e-03 [9.79e-04, 1.69e-03]	-0.02 [-0.13, +0.02]
$p_w = 1.6$ kPa RBDO-C	+0.7 ± 1.3	-0.5 ± 2.1	39/36	5	125/138	0/250	6.90e-04	6.25e-04 [5.35e-04, 8.05e-04]	+0.03 [-0.03, +0.04]

3.4. Monte Carlo convergence and sampling resolution

Fig. 5 shows the reliability indices of the base DDO and RBDO-C designs as functions of the sample size for five independent random streams. At the SAA sample size ($N = 5 \times 10^4$) the five streams give $\hat{\beta}_{ULS}$ between 2.97 and 3.16, i.e. a resolution of about ± 0.1 with which the optimiser can enforce $\beta_T = 3$; at $N = 10^6$ they agree within 3.02–3.05 (c.o.v. of $\hat{P}_f$ about 3%). The

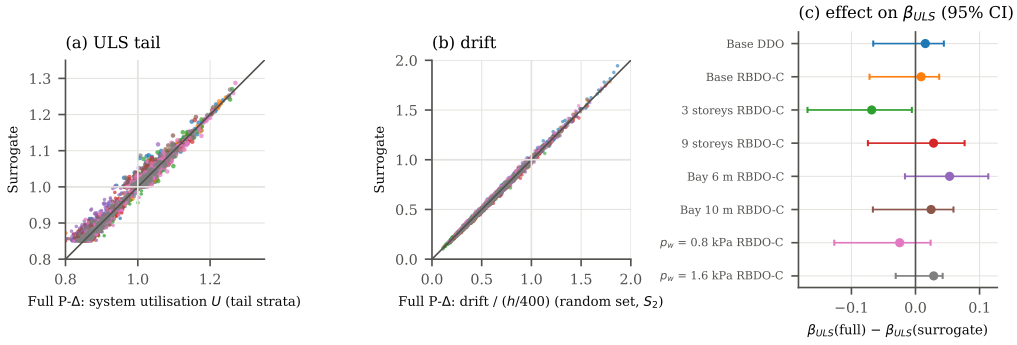


Figure 3: Surrogate versus full P-Δ re-analysis for eight designs: (a) system utilisation of the tail-stratified samples (large dots: stratum A, small dots: stratum B); (b) drift of the random sets; (c) difference of the ULS reliability index with 95% interval. Colours identify the designs as labelled in (c).

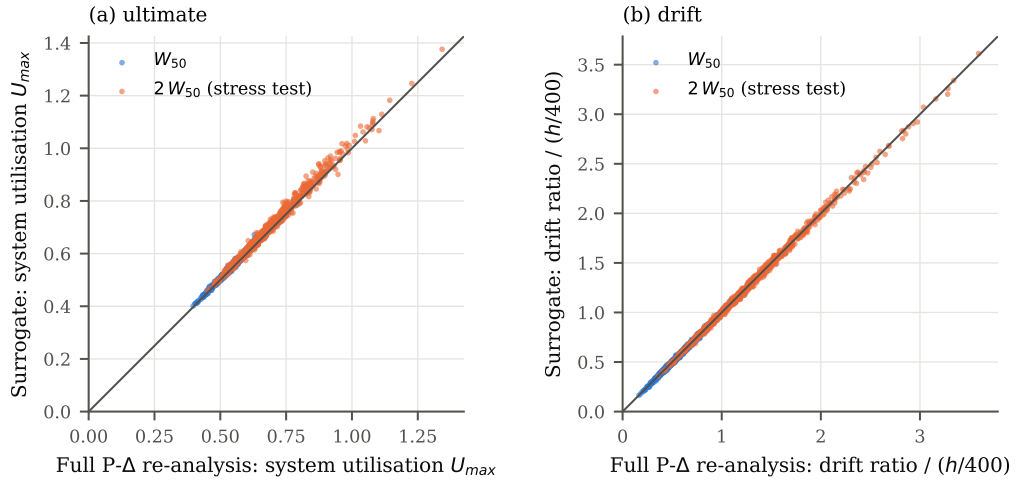


Figure 4: Surrogate versus full P-Δ re-analysis for the base RBDO-C design under the 50-year wind and under doubled wind (1000 samples each).

drift index converges much faster because its failure probability is larger. The SAA estimate therefore guides the search, and the independent 10^6-sample estimate with its confidence interval is the one reported.

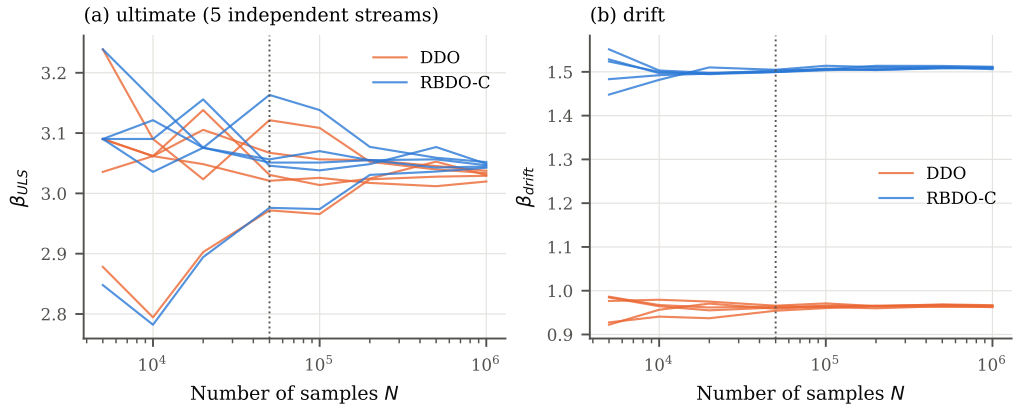


Figure 5: Convergence of the Monte Carlo reliability indices of the base designs (five independent streams each). The dotted line marks the SAA sample size used in the optimiser.

4. Results

4.1. Optimiser robustness and global-search credibility

Table 7 reports repeated runs. For the base frame, ten DDO runs gave the best mass of 18 261 kg in eight runs and 18 321 kg (+0.3%) in two, and five RBDO-C runs with different GA seeds all returned the same design (18 787 kg). For the other frames, two independent RBDO-C runs gave the same design in four of six frames; in the 6-m bay frame they differed by 0.1% and in the high-wind frame a third run was made because the first two differed: the three runs gave 21 738, 21 774 and 22 078 kg (spread 1.6%), the lightest coinciding with the design found by RBDO without code constraints. The GA with local search is thus robust within the tested budget for most frames, but not infallible; the reported optimum is always the lightest design found over all runs, and it is a local optimum with respect to the moves of the local search.

For the reduced problem (one section group, three variables, 11 200 designs), the exact optimum was obtained by evaluating the designs in order of

increasing mass until the first feasible one (Table 8). All ten GA runs of each formulation found the exact optimum, already before the local search, using about 700 of the 11 200 designs. The DDO optimum (21 289 kg) is preceded by 739 lighter, infeasible designs and the RBDO and RBDO-C optima (both 22 383 kg, +5.1%) by 960. Grouping all storeys into one section group also shows the value of the design grouping: the six-storey frame with three groups is 14% (DDO) and 16% (RBDO-C) lighter.

The SAA settings were varied for the base frame (Table 9). With $N = 2 \times 10^4$, 5×10^4 (three independent sample streams) and 10^5, RBDO-C returned the same design, whose SAA estimates of β_{ULS} ranged from 3.01 to 3.13 with the stream and size, while the independent verification gives 3.049 $[3.03, 3.07]$. Imposing the targets on the one-sided 95% lower confidence bounds in the SAA sample selected a heavier design (19 079 kg, +1.6%) with $\hat{\beta}_{ULS} = 3.125$ $[3.11, 3.14]$ and $\hat{\beta}_{drift} = 1.551$; this is the price of statistical conservatism at $N = 5 \times 10^4$ for this frame.

A DDO run took 2–12 min and an RBDO-C run 10–56 min (median 32 min) on one CPU core, the reliability evaluation accounting for a median of 83% of the RBDO-C time.

Table 7: Repeated optimisation runs (masses in kg). Distinct finals = number of different final designs; runs at best = runs that returned the lightest design. Designs eval. = mean number of distinct designs evaluated per run.

Problem	Formulation	Runs	Best (kg)	Median (kg)	Worst (kg)	Distinct finals	Runs at best	Designs eval.
3 storeys (9 var.)	DDO	2	7995	7995	7995	1	2	1556
3 storeys (9 var.)	RBDO-C	2	8188	8188	8188	1	2	1346
Base (9 var.)	DDO	10	18261	18261	18321	2	8	2130
Base (9 var.)	RBDO	2	18787	18787	18787	1	2	2074
Base (9 var.)	RBDO-C	5	18787	18787	18787	1	5	2109
9 storeys (9 var.)	DDO	2	32987	32987	32987	1	2	2247
9 storeys (9 var.)	RBDO-C	2	35450	35450	35450	1	2	1950
Bay 6 m (9 var.)	DDO	2	13707	13707	13707	1	2	1937
Bay 6 m (9 var.)	RBDO-C	2	14668	14674	14680	2	1	2220
Bay 10 m (9 var.)	DDO	2	25639	25639	25639	1	2	1938
Bay 10 m (9 var.)	RBDO-C	2	26065	26065	26065	1	2	2080
$p_w = 0.8$ kPa (9 var.)	DDO	2	17788	17788	17788	1	2	1999
$p_w = 0.8$ kPa (9 var.)	RBDO-C	2	17989	17989	17989	2	2	1979
$p_w = 1.6$ kPa (9 var.)	DDO	2	19743	19803	19862	2	1	2153
$p_w = 1.6$ kPa (9 var.)	RBDO-C	3	21738	21774	22078	3	1	2103
Reduced (3 var.)	DDO	10	21289	21289	21289	1	10	670
Reduced (3 var.)	RBDO	10	22383	22383	22383	1	10	673
Reduced (3 var.)	RBDO-C	10	22383	22383	22383	1	10	717

Table 8: Reduced problem (base frame with one section group): exact optimum by mass-ordered enumeration and success of ten GA runs per formulation (with and without the final local search).

Formulation	Designs	Exact optimum (kg)	Lighter designs (infeasible)	GA runs reaching it	GA without local search
DDO	11200	21289	739	10/10	10/10
RBDO	11200	22383	960	10/10	10/10
RBDO-C	11200	22383	960	10/10	10/10

Table 9: Base frame RBDO-C: effect of the SAA sample size, the SAA sample stream and the acceptance rule (point estimate or one-sided 95% lower confidence bound, LCB). Verified indices from 10^6 independent samples; vs. targets: a = above, w = within, b = below the target (95% interval), for ULS/drift.

SAA setting	Mass (kg)	SAA $\hat{\beta}_{ULS}$, $\hat{\beta}_{drift}$	k_{ULS}	Verified $\hat{\beta}_{ULS}$ [95% CI]	k_{drift}	Verified $\hat{\beta}_{drift}$ [95% CI]	vs. targets
$N = 20\,000$, stream A	18787	3.01, 1.51	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
$N = 50\,000$, stream A	18787	3.07, 1.50	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
$N = 50\,000$, stream C	18787	3.13, 1.51	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
$N = 50\,000$, stream B	18787	3.08, 1.51	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
$N = 100\,000$, stream A	18787	3.04, 1.50	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
$N = 50\,000$, stream A, 95% LCB	19079	3.13, 1.54	888	3.125 [3.11, 3.14]	60506	1.551 [1.55, 1.55]	a/a

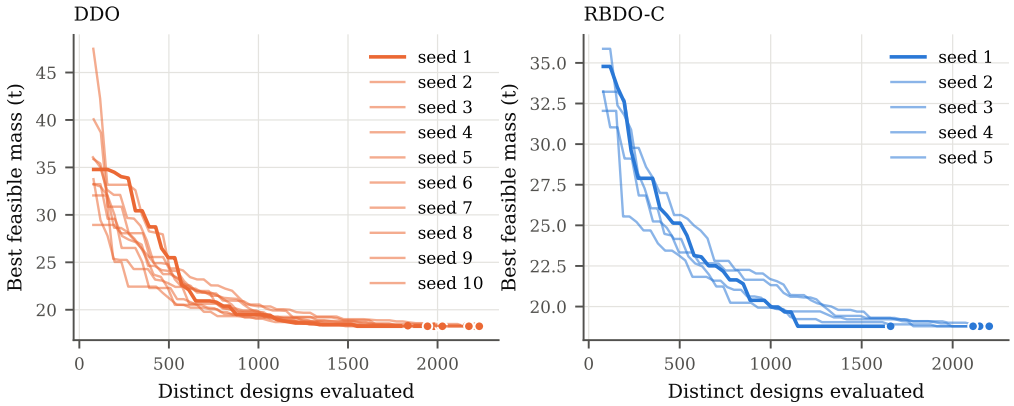


Figure 6: Convergence of the GA for the base frame (best feasible mass; ten DDO and five RBDO-C runs); dots mark the final designs after local search.

4.2. Deterministic versus code-compliant reliability-based design of the base frame

Table 10 compares the DDO and RBDO-C optima of the base frame. The DDO design weighs 18 261 kg; its governing LRFD ratio is 0.993 and its drift is 0.97 of the $h/400$ limit, so both strength and drift are active. Verified with 10^6 independent samples, it has $k = 1242$ ULS failures, $\hat{\beta}_{ULS} = 3.025$ $[3.01, 3.04]$, statistically above the target, but its 50-year drift reliability is only $\hat{\beta}_{drift} = 0.962$ ($\hat{P}_f = 0.17$): a drift check at the nominal 50-year wind leaves about a one-in-six chance that the drift limit is exceeded during the reference period. The reason is that the 50-year maximum wind has a mean-to-nominal ratio of 0.78 but a c.o.v. of 0.37, so the nominal wind is exceeded with a probability of about 0.19, and serviceability checks carry no load factor.

The RBDO-C design ($\beta_T = 3.0$, $\beta_{SLS} = 1.5$) weighs 18 787 kg (+2.88%), satisfies all LRFD checks (strength ratio 0.977) and has $\hat{\beta}_{ULS} = 3.049$ $[3.03, 3.07]$ ($k = 1148$) and $\hat{\beta}_{drift} = 1.507$ $[1.50, 1.51]$; both reliability constraints are active and both targets are met with 95% confidence. It is the same design as found by RBDO without code constraints, so for the base frame the code checks are not binding once the reliability targets are imposed. The additional steel is placed where it buys stiffness most cheaply: the lower-storey beams are deepened (W21×50 to W24×55) and the interior columns of storeys 3–4 increased; its deterministic drift ratio is 0.80 of $h/400$, i.e. $h/500$. The exceedance curves of Fig. 7 show that the two designs have virtually identical distributions of the system utilisation, whereas the drift distribution of the RBDO-C design is shifted to the left over its whole range.

Fig. 8 relates the member utilisations of the deterministic check to their probabilistic counterparts. The 99.9% quantile of the member utilisation lies 5–15% below the LRFD ratio for most members, which reflects the margin built into the partial factors. In 96% of the failed samples of both designs the governing member is a beam (Table 14), mostly the W21×44 beams at floor levels 3–5. The gravity scenario S_1 alone causes 79% (DDO) and 95% (RBDO-C) of the failures; stiffening the frame against drift removes most wind-scenario failures without changing the gravity-governed ULS reliability.

4.3. Cost–reliability trade-off

Fig. 9 shows the non-dominated designs of the bi-objective problem (mass versus β_{ULS}, subject to $\beta_{drift}, \beta_{defl} \geq 1.5$), re-verified with 4×10^5 independent samples and shown with their 95% intervals, together with all feasible designs visited by NSGA-II (SAA estimates). The lightest front design (18 747 kg)

Table 10: Optimum designs of the base frame and their deterministic and probabilistic performance (10^6 independent samples; exact 95% confidence intervals; for zero failures the one-sided 95% lower confidence bound of β is given).

	DDO			RBDO-C		
Storeys	Ext. col.	Int. col.	Beam	Ext. col.	Int. col.	Beam
1-2	W14X61	W14X90	W21X50	W14X61	W14X90	W24X55
3-4	W14X48	W14X61	W21X44	W14X48	W14X68	W21X44
5-6	W14X48	W14X48	W21X44	W14X48	W14X48	W21X44
Steel mass (kg)	18261			18787		
Distinct sections	5			6		
LRFD strength ratio	0.993			0.977		
Drift ratio / limit (D+0.5L+W)	0.972			0.798		
Deflection / limit (L)	0.156			0.156		
Max. B_2 (DAM)	1.186			1.151		
ULS failures k in $N = 10^6$	1242			1148		
$\hat{P}_{f,ULS}$ [95% CI]	1.242e-03 [1.174e-03, 1.313e-03]			1.148e-03 [1.083e-03, 1.216e-03]		
$\hat{\beta}_{ULS}$ [95% CI]	3.025 [3.01, 3.04]			3.049 [3.03, 3.07]		
Drift failures k	167961			65968		
$\hat{\beta}_{drift}$ [95% CI]	0.962 [0.96, 0.97]			1.507 [1.50, 1.51]		
Deflection failures k; β_{defl}	0; ≥4.53			0; ≥4.53		

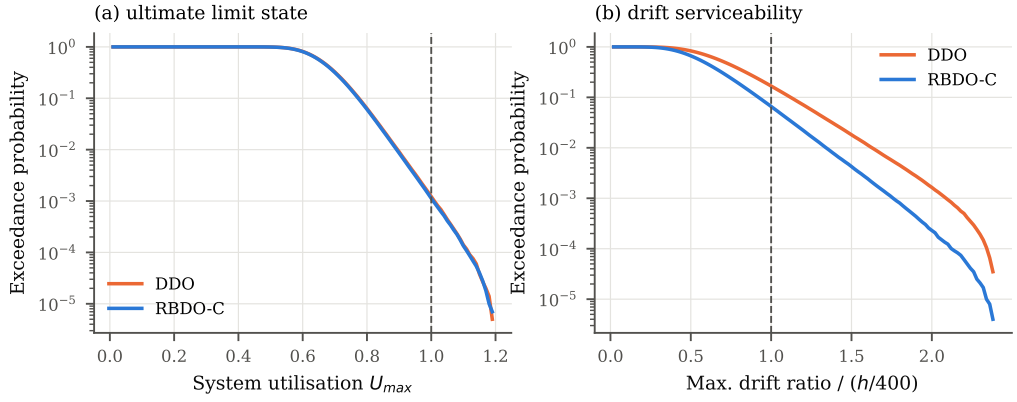


Figure 7: Exceedance probabilities of the system utilisation (ULS) and of the maximum inter-storey drift (normalised by $h/400$) of the base DDO and RBDO-C designs over 50 years (10^6 samples).

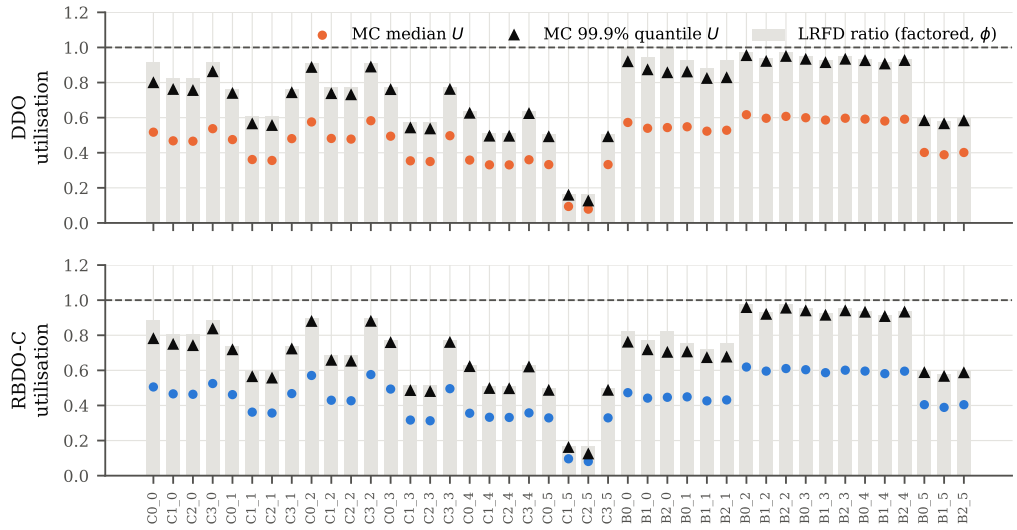


Figure 8: Member utilisations of the base designs: governing LRFD ratio (bars) and median and 99.9% quantile of the probabilistic utilisation. Member labels: Ci_s is the column on line i in storey s, Bi_s the beam in bay i at floor s (counted from 0).

fails the LRFD strength check (ratio 1.10) and has $\hat{\beta}_{ULS} = 2.36$; all other front designs are code-compliant. Because drift governs the mass once $\beta_{SLS} = 1.5$ is imposed, the code-compliant front is steep: raising $\hat{\beta}_{ULS}$ from 3.06 to 3.36, 3.64 and 3.83 costs +1.3%, +3.5% and +4.2% of steel (19 039, 19 452 and 19 568 kg against 18 787 kg). All front designs have $\hat{\beta}_{drift}$ between 1.51 and 1.57. The verified indices of the high-reliability front designs are lower than their SAA estimates (3.83 against 4.26 for the heaviest), which illustrates the SAA optimisation bias at small failure counts: the SAA sample contains only a few failures there, and the optimiser selects designs for which the particular sample is favourable. The front passes through the RBDO-C optimum, which is an independent check of both optimisers. The DDO optimum lies to the left of the front but is infeasible for it because $\hat{\beta}_{drift} = 0.96$.

4.4. Effect of the reliability targets and of the deterministic drift limit

Table 11 isolates the role of the targets for the base frame. With $\beta_T = 3.0$ and $\beta_{SLS} = 0.5$, RBDO returns *exactly* the DDO design: the code-based optimum is also the reliability-based optimum when the reliability levels it implicitly provides are required. Raising β_{SLS} to 1.0 and 1.5 costs +0.97%

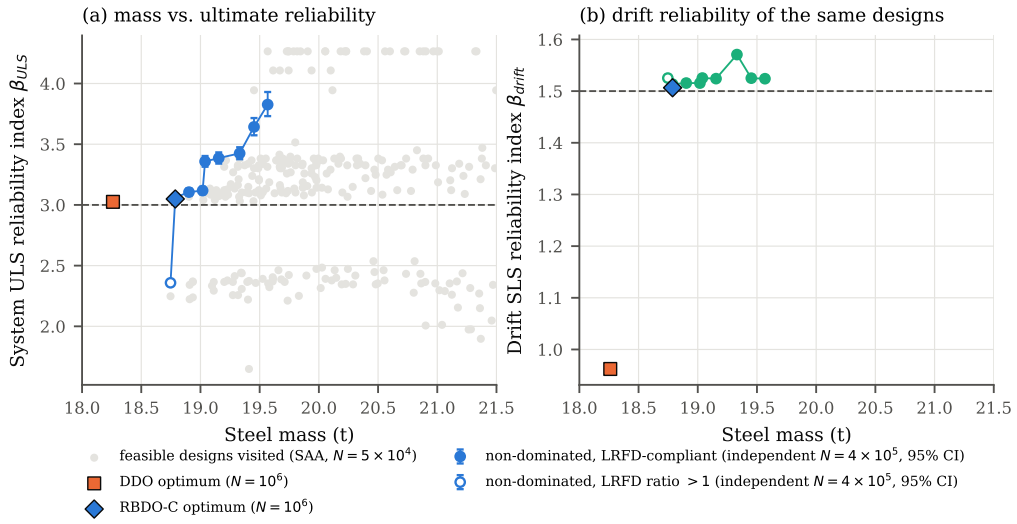


Figure 9: Cost–reliability trade-off of the base frame. (a) Non-dominated designs of the bi-objective problem, re-verified with 4×10^5 independent samples (95% intervals; open marker: fails the LRFD strength check), and all feasible designs visited by NSGA-II (SAA estimates, $N = 5 \times 10^4$; values at the top are capped by the sample size); (b) drift reliability of the same non-dominated designs. The DDO and RBDO-C optima (10^6 samples) are shown for reference.

and +2.88% of steel, and raising β_T to 3.5 at $\beta_{SLS} = 1.5$ costs +6.52% ($\hat{\beta}_{ULS} = 3.67$, above the target because of the discreteness of the catalogue).

Table 12 and Fig. 10 compare RBDO-C with deterministic optima for drift limits of $h/400$, $h/450$, $h/500$ and $h/600$, all with the same strength and deflection checks. Tightening the drift limit raises β_{drift} of the drift-governed frames in a similar way (about 1.2–1.4 at $h/450$, 1.5 at $h/500$ and 2.0 at $h/600$), but it has no effect on frames governed by strength (three storeys, low wind) and cannot raise their β_{ULS}. Applying the acceptance rule of Section 2.5 to the verified indices, the lightest deterministic design that meets both targets is $h/500$ for the base, nine-storey, 6-m bay and high-wind frames and $h/600$ for the 10-m bay and low-wind frames; for the three-storey frame no drift limit meets β_T. Compared with that design, RBDO-C is equal (base), lighter by 0.10% (nine storeys), 0.34% (low wind), 1.14% (high wind) and 4.99% (10-m bays), and heavier by 0.58% for the 6-m bays. The last case is instructive: the $h/500$ design has $\hat{\beta}_{drift} = 1.499$ $[1.495, 1.503]$, statistically indistinguishable from the target and therefore accepted in verification, but its SAA estimate (1.494) was below the target, so the point-estimate SAA constraint rejected it. Near-threshold classification can thus change the optimum by about 0.5% of mass in either direction. No single drift limit delivers the target reliabilities in all frames, which is what the explicit reliability constraints achieve by construction.

Table 11: Base frame: optimum mass for different reliability targets and drift limits (10^6 independent samples; Δm from unrounded masses relative to DDO with $h/400$).

Formulation	β_T	β_{SLS}	Mass (kg)	Δm	k_{ULS}	$\hat{\beta}_{ULS}$ [95% CI]	$\hat{\beta}_{drift}$	LRFD
DDO, drift $h/400$	–	–	18261	+0.00%	1242	3.025 [3.01, 3.04]	0.962	0.993
DDO, drift $h/450$	–	–	18626	+2.00%	1039	3.079 [3.06, 3.10]	1.410	0.973
DDO, drift $h/500$	–	–	18787	+2.88%	1148	3.049 [3.03, 3.07]	1.507	0.977
DDO, drift $h/600$	–	–	20664	+13.16%	810	3.152 [3.13, 3.17]	2.009	0.949
RBDO	3.0	1.5	18787	+2.88%	1148	3.049 [3.03, 3.07]	1.507	0.977
RBDO-C	3.0	1.5	18787	+2.88%	1148	3.049 [3.03, 3.07]	1.507	0.977
RBDO	3.0	0.5	18261	+0.00%	1242	3.025 [3.01, 3.04]	0.962	0.993
RBDO	3.0	1.0	18438	+0.97%	1113	3.058 [3.04, 3.08]	1.005	0.982
RBDO	3.5	1.5	19452	+6.52%	120	3.673 [3.63, 3.72]	1.518	0.894

4.5. Parametric study and statistical verification of all final designs

Table 13 reports, for every frame, the DDO, RBDO and RBDO-C optima with their masses, LRFD ratios, failure counts and 95% intervals, and Fig. 11 shows the same results graphically. Four observations stand out.

Table 12: Optimum mass (kg) and verified $\hat{\beta}_{ULS}/\hat{\beta}_{drift}$ of deterministic designs with four drift limits and of RBDO-C. Bold: meets both targets under the acceptance rule (target not below the 95% interval). Last column: RBDO-C relative to the lightest deterministic design meeting both targets.

Frame	DDO $h/400$	DDO $h/450$	DDO $h/500$	DDO $h/600$	RBDO-C	RBDO-C vs. lightest DDO meeting targets
3 storeys	7995 (2.93/2.54)	7995 (2.93/2.54)	7995 (2.93/2.54)	7995 (2.93/2.54)	8188 (3.00/2.71)	–
Base	18261 (3.03/0.96)	18626 (3.08/1.41)	**18787 (3.05/1.51)**	**20664 (3.15/2.01)**	18787 (3.05/1.51)	+0.00%
9 storeys	32987 (3.27/0.90)	34665 (3.29/1.28)	**35484 (3.18/1.53)**	**38784 (3.16/2.05)**	35450 (3.35/1.53)	-0.10%
Bay 6 m	13707 (3.66/0.95)	14140 (3.44/1.31)	**14583 (3.76/1.50)**	**15960 (4.12/2.03)**	14668 (3.73/1.55)	+0.58%
Bay 10 m	25639 (2.88/1.15)	26035 (2.93/1.22)	26035 (2.92/1.66)	**27435 (3.15/2.02)**	26065 (3.12/1.66)	-4.99%
$p_w = 0.8$ kPa	17788 (2.92/1.78)	17788 (2.92/1.78)	17788 (2.92/1.78)	**18050 (3.01/2.02)**	17989 (3.02/1.61)	-0.34%
$p_w = 1.6$ kPa	19743 (3.11/0.89)	20679 (3.37/1.21)	**21989 (3.13/1.53)**	**23385 (3.36/2.02)**	21738 (3.19/1.51)	-1.14%

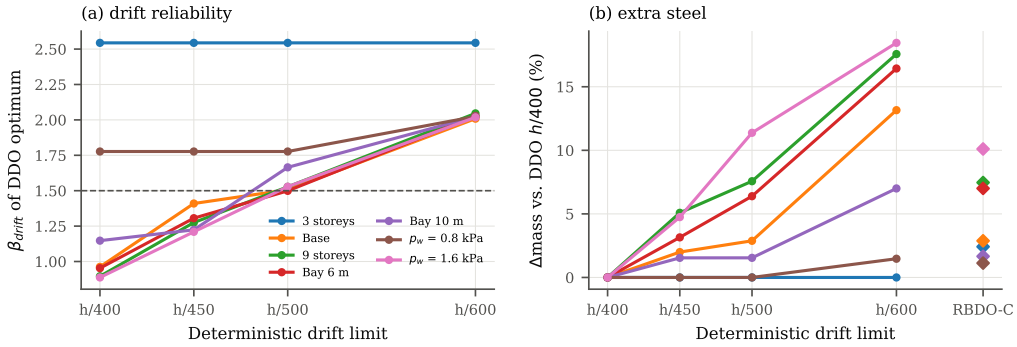


Figure 10: Deterministic optima with tightened drift limits: (a) verified drift reliability index; (b) mass increase relative to DDO with $h/400$, with the RBDO-C designs (diamonds) for comparison.

First, the reliability of deterministic optima is far from uniform. Although every DDO design has a governing strength or drift check within 7% of its limit, the verified system indices range from $\hat{\beta}_{ULS} = 2.880$ $[2.87, 2.89]$ (10-m bays) to 3.656 $[3.61, 3.70]$ (6-m bays), and the drift indices from 0.887 (high wind) to 2.544 (three storeys). Three DDO designs fall significantly below $\beta_T = 3.0$ (three storeys, 10-m bays, low wind) and five have $\hat{\beta}_{drift} < 1.5$, four of them below 1.0.

Second, RBDO-C equalises the reliability. All seven RBDO-C designs satisfy every LRFD check and meet both targets under the acceptance rule: six are above both targets with 95% confidence, and the three-storey design has $\hat{\beta}_{ULS} = 2.995$ $[2.98, 3.01]$, within the interval of the target. The mass premium relative to DDO, computed from unrounded masses, ranges from +1.13% (low wind) to +10.11% (high wind). The premium is essentially the cost of raising β_{drift} to 1.5 in the drift-governed frames and of raising β_{ULS} to 3.0 in the strength-governed ones.

Third, the code checks matter in two of seven frames. Without them, RBDO selects a 6-m bay design with an LRFD strength ratio of 1.011 and a low-wind design with 1.002, i.e. designs that meet the reliability targets but would marginally fail the mandatory strength check; in the 6-m bay frame RBDO trades the surplus ULS reliability of the DDO design ($\hat{\beta}_{ULS} = 3.66$) for stiffness ($\hat{\beta}_{ULS} = 3.22$). Adding the code checks (RBDO-C) raises the mass by 0.32% (6-m bays) and 0.64% (low wind) relative to RBDO. In the other five frames the RBDO optimum already satisfies the code checks and RBDO-C returns the same design. The conclusions on mass are therefore unchanged when all RBDO candidates must satisfy the deterministic LRFD checks; the premiums over DDO change from +6.67% to +7.01% (6-m bays) and from +0.48% to +1.13% (low wind).

Fourth, the location of the extra steel explains the premiums (Tables 14 and 15). In the drift-governed frames (base, nine storeys, high wind) RBDO-C adds steel to the beams of the lower storey groups and to the interior columns, which stiffens the frame; in the high-wind frame this adds 755 kg to columns and 1 240 kg to beams. In the 10-m bay frame, where DDO falls below β_T because the long beams are governed by the 50-year live load (95% of failures in S_1, 95% in beams), RBDO-C deepens the beams (+577 kg) and lightens the columns (−151 kg), after which the remaining failures are shared between beams (57%) and columns (43%). In the 6-m bay frame, governed by drift but with short, lightly loaded beams, failures occur mostly in interior columns (77–79%) in the wind scenario.

Table 13: Statistical verification of all final designs ($N = 10^6$ independent samples). Δm: mass relative to DDO (unrounded); LRFD: governing strength ratio; k: number of failures; vs. targets: a = above, w = within, b = below the target, by the exact 95% interval, for ULS/drift. No deflection failure occurred in any design ($\beta_{defl} \geq 4.53$ with 95% confidence).

Frame	Formulation	Mass (kg)	Δm	LRFD	k_{ULS}	$\hat{\beta}_{ULS}$ [95% CI]	k_{drift}	$\hat{\beta}_{drift}$ [95% CI]	vs. targets
3 storeys	DDO	7995	–	0.991	1707	2.928 [2.91, 2.94]	5481	2.544 [2.53, 2.55]	b/a
	RBDO	8188	+2.42%	0.978	1370	2.995 [2.98, 3.01]	3357	2.711 [2.70, 2.72]	w/a
	RBDO-C	8188	+2.42%	0.978	1370	2.995 [2.98, 3.01]	3357	2.711 [2.70, 2.72]	w/a
Base	DDO	18261	–	0.993	1242	3.025 [3.01, 3.04]	167961	0.962 [0.96, 0.97]	a/b
	RBDO	18787	+2.88%	0.977	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
	RBDO-C	18787	+2.88%	0.977	1148	3.049 [3.03, 3.07]	65968	1.507 [1.50, 1.51]	a/a
9 storeys	DDO	32987	–	0.987	535	3.271 [3.25, 3.30]	185169	0.896 [0.89, 0.90]	a/b
	RBDO	35450	+7.47%	0.965	403	3.351 [3.32, 3.38]	62958	1.530 [1.53, 1.53]	a/a
	RBDO-C	35450	+7.47%	0.965	403	3.351 [3.32, 3.38]	62958	1.530 [1.53, 1.53]	a/a
Bay 6 m	DDO	13707	–	0.936	128	3.656 [3.61, 3.70]	170283	0.953 [0.95, 0.96]	a/b
	RBDO	14621	+6.67%	1.011	642	3.220 [3.20, 3.24]	60925	1.547 [1.54, 1.55]	a/a
	RBDO-C	14668	+7.01%	0.914	95	3.732 [3.68, 3.78]	60850	1.548 [1.54, 1.55]	a/a
Bay 10 m	DDO	25639	–	0.993	1986	2.880 [2.87, 2.89]	125645	1.147 [1.14, 1.15]	b/b
	RBDO	26065	+1.66%	0.953	906	3.119 [3.10, 3.14]	47974	1.665 [1.66, 1.67]	a/a
	RBDO-C	26065	+1.66%	0.953	906	3.119 [3.10, 3.14]	47974	1.665 [1.66, 1.67]	a/a
$p_w = 0.8$ kPa	DDO	17788	–	0.999	1760	2.918 [2.90, 2.93]	37801	1.777 [1.77, 1.78]	b/a
	RBDO	17874	+0.48%	1.002	1415	2.986 [2.97, 3.00]	23753	1.982 [1.98, 1.99]	w/a
	RBDO-C	17989	+1.13%	0.998	1267	3.019 [3.00, 3.04]	53870	1.608 [1.60, 1.61]	a/a
$p_w = 1.6$ kPa	DDO	19743	–	0.995	940	3.109 [3.09, 3.13]	187435	0.887 [0.88, 0.89]	a/b
	RBDO	21738	+10.11%	0.963	714	3.189 [3.17, 3.21]	65403	1.511 [1.51, 1.51]	a/a
	RBDO-C	21738	+10.11%	0.963	714	3.189 [3.17, 3.21]	65403	1.511 [1.51, 1.51]	a/a

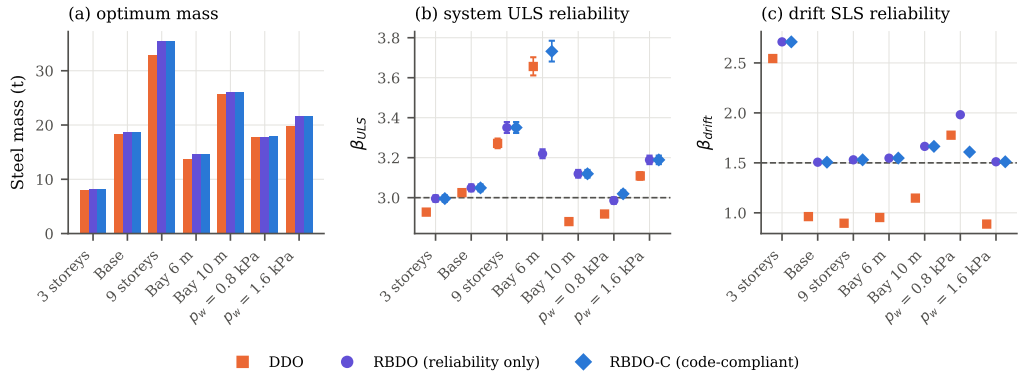


Figure 11: Parametric study: (a) optimum steel mass, (b) system ULS and (c) drift reliability indices with exact 95% intervals of the DDO, RBDO and RBDO-C designs (10^6 independent samples); dashed lines mark the targets.

Table 14: Where ULS failures occur (10^6 samples): share of failures by governing member type and by scenario (S_1: gravity with L_{50}; S_2: wind).

Frame	Design	k_{ULS}	Beams (%)	Ext. col. (%)	Int. col. (%)	S_1 only / S_2 only / both (%)
3 storeys	DDO	1707	67	0	33	94 / 6 / 0
	RBDO-C	1370	67	0	33	97 / 3 / 0
Base	DDO	1242	96	4	0	79 / 21 / 0
	RBDO-C	1148	96	4	0	95 / 5 / 0
9 storeys	DDO	535	35	1	63	60 / 40 / 0
	RBDO-C	403	65	1	34	70 / 30 / 0
Bay 6 m	DDO	128	15	8	77	14 / 86 / 0
	RBDO-C	95	16	5	79	25 / 75 / 0
Bay 10 m	DDO	1986	95	1	4	95 / 5 / 0
	RBDO-C	906	57	29	14	93 / 7 / 0
$p_w = 0.8$ kPa	DDO	1760	60	2	38	94 / 6 / 0
	RBDO-C	1267	32	0	68	98 / 2 / 0
$p_w = 1.6$ kPa	DDO	940	89	8	2	64 / 36 / 0
	RBDO-C	714	97	3	0	97 / 3 / 0

4.6. Sections and an illustrative cost model

Table 15 lists the number of distinct sections of the DDO and RBDO-C designs, the change of column and beam mass, and the cost premium of RBDO-C under the cost index (11) for τ between 0.25 and 1 t of steel per distinct section and κ of 0 or 0.25. RBDO-C uses one section more than DDO in three frames, one fewer in two and the same number in two. Accordingly, the non-mass terms increase the premium for the base, low- and high-wind frames (base: from 2.9% to 4.9% for $\tau = 0.5$ t) and reduce it for the nine-storey and 6-m bay frames; only for the 6-m bays with $\tau = 1$ t and $\kappa = 0$ does RBDO-C become marginally cheaper (−0.2%). Within this range of cost ratios, the ranking of DDO and RBDO-C does not change in any other case, so the conclusions drawn from steel mass carry over to this broader, although uncalibrated, cost measure.

Table 15: Distinct sections, change of column and beam mass, and cost premium of RBDO-C over DDO under the illustrative cost index $C = m + \tau n_{sec} + \kappa m_{beam}$ (τ in tonnes of steel per distinct section).

	Distinct sections		Δm RBDO-C vs. DDO (kg)			Cost premium of RBDO-C over DDO				
Frame	DDO	RBDO-C	Columns	Beams	Mass	$\tau = 0.25$, $\kappa = 0.0$	$\tau = 0.5$, $\kappa = 0.0$	$\tau = 1.0$, $\kappa = 0.0$	$\tau = 0.5$, $\kappa = 0.25$	$\tau = 1.0$, $\kappa = 0.25$
3 storeys	4	4	+193	+0	+2.42%	+2.15%	+1.93%	+1.61%	+1.74%	+1.48%
Base	5	6	+162	+365	+2.88%	+3.98%	+4.94%	+6.56%	+4.81%	+6.29%
9 storeys	8	7	+785	+1677	+7.47%	+6.32%	+5.31%	+3.57%	+5.74%	+4.14%
Bay 6 m	7	6	-115	+1076	+7.01%	+4.59%	+2.67%	-0.19%	+3.90%	+1.03%
Bay 10 m	6	6	-151	+577	+1.66%	+1.57%	+1.49%	+1.35%	+1.74%	+1.59%
$p_w = 0.8$ kPa	5	6	-212	+413	+1.13%	+2.37%	+3.45%	+5.27%	+3.55%	+5.18%
$p_w = 1.6$ kPa	6	7	+755	+1240	+10.11%	+10.57%	+10.97%	+11.63%	+11.03%	+11.63%

4.7. Sensitivity to the probabilistic model and to model uncertainty

Fig. 12 shows how the verified reliability indices of the base designs change when one assumption of Table 3 is altered (4×10^5 samples each). Among the physical variables, the system ULS index is most sensitive to the live-load model (c.o.v. 0.35 instead of 0.25: $\Delta\beta \approx -0.5$), the fabrication factor (removing it: $+0.3$) and the yield-stress model (mean-to-nominal 1.05 instead of 1.10, or c.o.v. 0.10 instead of 0.07: -0.2). Inter-member correlation of F_y changes β_{ULS} by ± 0.05, and the out-of-plumbness and modulus variability are negligible. The drift index depends almost exclusively on the wind model: a mean-to-nominal ratio of 0.90 instead of 0.78 lowers it by 0.4, and the c.o.v. changes it by about ± 0.2. The DDO and RBDO-C designs react in the same way, so the comparison between them is robust to these assumptions.

Model uncertainty has a larger effect on the absolute ULS indices (Table 16). Frame-wide model factors on resistance, load effect and drift with c.o.v. 0.05 lower β_{ULS} by 0.23–0.30 and with c.o.v. 0.10 by 0.82–0.90 in all frames, for DDO and RBDO-C alike, whereas β_{drift} changes by at most 0.05 because the drift uncertainty is dominated by the wind. The baseline indices of this paper are therefore conditional on the structural and resistance models: with a model uncertainty of 0.10, a design meeting $\beta_T = 3.0$ in the baseline model would have an index of about 2.1–2.3. Because the shift is almost uniform across frames and formulations, the *relative* comparison of DDO and RBDO-C is not affected; a target that is to be met including model uncertainty would have to be raised by the corresponding amount, at the mass cost shown in Fig. 9.

5. Discussion

Answer to the research question. For the frames studied, uncertainty-aware optimisation did *not* produce lighter designs than deterministic optimisation against the LRFD checks; the code-compliant reliability-based designs were 1.1–10.1% heavier. The deterministic optima are already close to the reliability-based optimum for the ULS ($\hat{\beta}_{ULS} \approx 2.9$–$3.7$ at a target of 3.0), while the conventional drift check at the nominal 50-year wind provides only $\hat{\beta}_{drift} \approx 0.9$–$1.2$ for the drift-governed frames. When the reliability levels implicit in the code are required, RBDO returns the DDO design exactly (Table 11). RBDO therefore provides economy *at a prescribed and uniform reliability*, not with respect to code-based design, whose reliability is unknown and non-uniform. Against deterministic designs with a tightened drift limit that

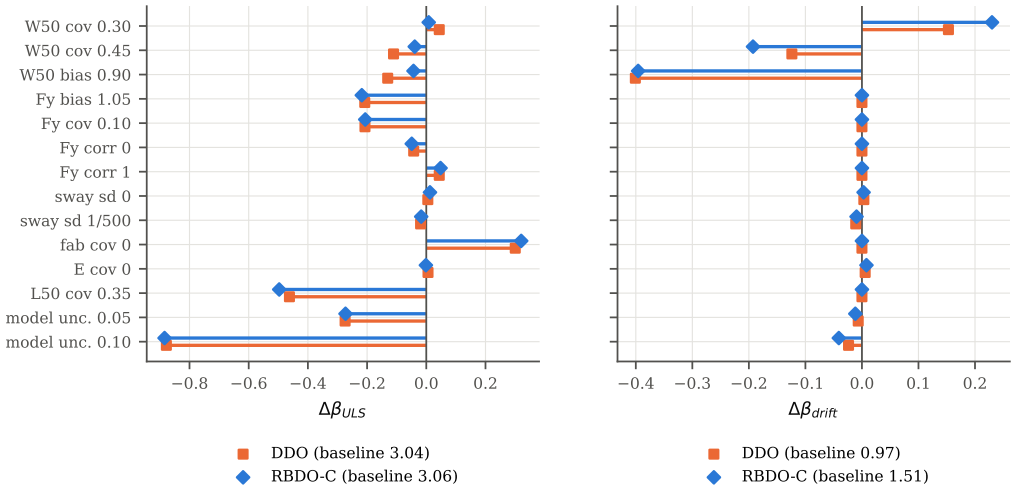


Figure 12: Change of the reliability indices of the base designs when one assumption of the probabilistic model is changed, including frame-wide model uncertainty factors (baseline indices in the legend).

Table 16: Verified reliability indices of the final designs without and with frame-wide model uncertainty factors θ_R, θ_E and θ_δ (lognormal, unit mean, common c.o.v. 0.05 or 0.10; 10^6 samples).

		$\hat{\beta}_{ULS}$			$\hat{\beta}_{drift}$		
Frame	Design	none	0.05	0.10	none	0.05	0.10
3 storeys	DDO	2.93	2.64	2.04	2.54	2.53	2.47
	RBDO-C	3.00	2.70	2.09	2.71	2.69	2.62
Base	DDO	3.03	2.76	2.17	0.96	0.96	0.94
	RBDO-C	3.05	2.77	2.19	1.51	1.50	1.47
9 storeys	DDO	3.27	2.98	2.37	0.90	0.89	0.88
	RBDO-C	3.35	3.07	2.45	1.53	1.52	1.49
Bay 6 m	DDO	3.66	3.43	2.84	0.95	0.95	0.93
	RBDO-C	3.73	3.46	2.83	1.55	1.54	1.50
Bay 10 m	DDO	2.88	2.60	2.04	1.15	1.14	1.12
	RBDO-C	3.12	2.83	2.22	1.66	1.65	1.62
$p_w = 0.8$ kPa	DDO	2.92	2.63	2.04	1.78	1.76	1.73
	RBDO-C	3.02	2.73	2.12	1.61	1.60	1.56
$p_w = 1.6$ kPa	DDO	3.11	2.84	2.26	0.89	0.88	0.87
	RBDO-C	3.19	2.92	2.33	1.51	1.50	1.47

meet the same targets, RBDO-C was lighter by up to 5.0% in five frames, equal in one and 0.6% heavier in one, the last because of near-threshold sampling classification. These conclusions agree in sign with the observation of Papadrakakis et al. (2004) that a reliability-based steel-frame design was heavier than its deterministic counterpart.

What the reliability indices mean. β_{ULS} is the index of the first attainment of a code-format member limit state in a second-order elastic analysis (Section 2.3). It is consistent with the component-based calibration of the LRFD factors, which is what makes the comparison with the code meaningful, but it is neither a collapse reliability nor an absolute measure of safety. Three omissions act in opposite directions. Inelastic redistribution in these redundant frames would raise the collapse reliability (Buonopane and Schafer, 2006; Zhang et al., 2016). Residual stresses and member imperfections are represented only through the AISC column and LTB curves, not through an explicit imperfect model, and connection behaviour is not modelled. Model uncertainty, if included, lowers the indices by up to about 0.9 for a c.o.v. of 0.10 (Table 16). The validation against full P-Δ re-analysis (Section 3.3) removes the surrogate approximations but not these model-form choices. The reported indices should therefore be read as conditional, comparative measures, appropriate for comparing designs and formulations under one consistent model, and not as estimates of real-world failure rates.

Statistical confidence and acceptance. With $N = 10^6$ independent samples and 400–2000 ULS failures per final design, the 95% intervals of β_{ULS} are about ± 0.02–0.05, and of β_{drift} about ± 0.005. The reliability constraints are imposed on SAA point estimates with $N = 5 \times 10^4$, whose resolution is about ± 0.1 in β_{ULS}; designs found by the optimiser therefore lie up to about 0.1 on either side of the target in verification, and three of 21 final designs (Table 13) are statistically indistinguishable from the target. Requiring the SAA lower confidence bound to meet the target removes this ambiguity at a cost of 1.6% of steel for the base frame (Table 9). Which rule is appropriate depends on whether the target is understood as a point value (as in code calibration) or as a value to be demonstrated with confidence.

Cost of reliability. The trade-off curves are strongly asymmetric. Once a frame is drift-governed, its ULS reliability can be raised from 3.0 to about 3.8 for 4% more steel (Fig. 9), whereas the serviceability index is expensive to raise (2.9% of steel from $\beta_{drift} = 0.96$ to 1.51 in the base frame and 10.1%

from 0.89 to 1.51 in the high-wind frame). Because the ULS and SLS indices of a design are governed by different scenarios and random variables (Table 14, Fig. 12), a single global safety measure would hide these trade-offs.

What has and has not been demonstrated. Demonstrated: for regular planar steel moment frames of three to nine storeys with 6–10-m bays under gravity and wind, with discrete W-shapes and the stated probabilistic model, (i) the superposition surrogate reproduces full second-order Monte Carlo results within about ± 0.07 in β_{ULS}; (ii) the GA with local search finds the exact optimum of a reduced problem in all runs and repeats its full-problem results in most frames; and (iii) code-compliant RBDO costs 1–10% more steel than DDO and is generally the least-mass way to meet both targets. Not demonstrated: three-dimensional frames and torsion, multiple wind directions, seismic actions, inelastic collapse reliability, connection and foundation design, cost models calibrated to fabrication practice, and live-load models with spatial variability. A three-objective NSGA-II run (mass, β_{ULS}, β_{drift}) with about 3500 designs did not approach the single-objective optima (lightest design with both targets 19 500 kg against 18 787 kg), so for discrete frame problems of this size an ε-constraint approach with a strong single-objective optimiser, complemented by a bi-objective front, is the more economical way to map the trade-off.

6. Conclusions

A reproducible framework for reliability-constrained optimisation of steel moment frames was developed on the open-source PyNite library and applied to seven planar frames with discrete W-shape sections. The main conclusions are:

1. Five linear analyses per design, combined by superposition with AISC B_1–B_2 amplification, give a reliability surrogate that agrees with OpenSeesPy to machine precision for linear responses and within 2% for P-Δ responses. Against full P-Δ re-analysis with member-by-member random stiffness, tail-stratified for eight designs, it reproduces β_{ULS} within -0.07 to $+0.05$ and the drift reliability without detectable bias.
2. Deterministic minimum-mass designs that satisfy the LRFD checks have non-uniform reliability: $\hat{\beta}_{ULS}$ between 2.88 and 3.66, three of seven significantly below 3.0, and 50-year $\hat{\beta}_{drift}$ between 0.89 and 2.54.

3. Code-compliant RBDO with $\beta_T = 3.0$ and $\beta_{SLS} = 1.5$ met both targets in independent 10^6-sample verification (six designs with 95% confidence, one within the interval of the target) and was 1.1–10.1% heavier than DDO. Without the code checks, RBDO selected two designs that marginally fail the LRFD strength check; enforcing the checks added 0.3–0.6% of steel and did not change any conclusion. With code-implied targets, RBDO reproduces the DDO design.
4. Uncertainty-aware optimisation therefore did not reduce steel mass relative to code-based optimisation; it delivered designs with prescribed, uniform reliability, generally at less steel than tightened deterministic drift limits that meet the same targets. These conclusions hold under an illustrative cost model with section-count and connection terms.
5. The absolute indices are conditional on the structural model and the load statistics: model uncertainty with c.o.v. 0.10 lowers β_{ULS} by about 0.9, and the live- and wind-load models change the indices by up to 0.5; the relative comparison of the formulations is insensitive to these choices.

Data and code availability

The framework, the input data, the scripts that produce every figure and table, and the raw results are openly available at https://github.com/karthice1/Pynite in the folder `Research/reliability_optimization` (branch `claude/project-thread-o7hjbo`; the commit used for this manuscript is recorded in the folder's `README.md`). The software environment is listed in `requirements.txt` (Python 3.13, NumPy, SciPy, Matplotlib, OpenSeesPy 3.8 and PyNite from the same repository). Every stochastic computation uses fixed seeds that are stored with its results. The single command `./reproduce.sh` reruns all optimisations, verifications and validations, and regenerates every figure, table and this manuscript; the stages can also be run individually with `run_experiments.py`. A persistent archive (release tag and DOI) will be created on acceptance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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